

Robustness to Data Distribution Shifts: Test-Time Adaptation Using TENT

Presented by Daniel Gulvin, CSE 710, 10/8/2025

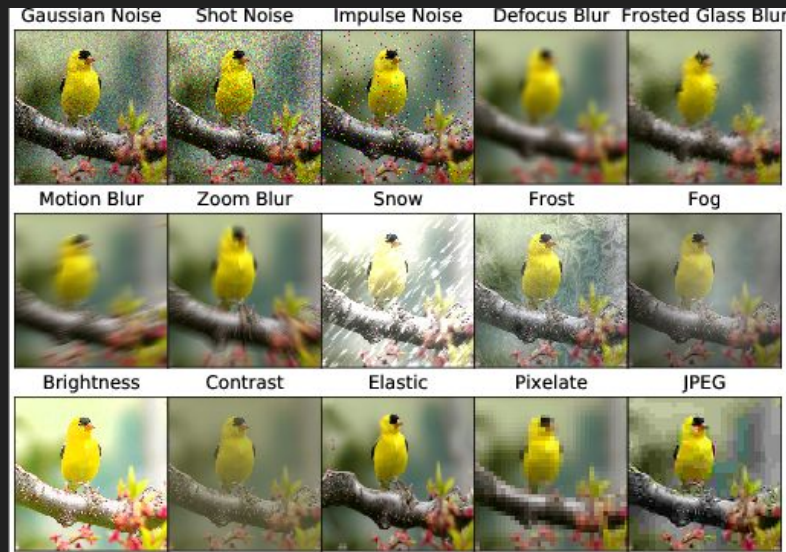
Distribution Shifts

- Distribution shift = mismatch between training and test data
- Causes
 - Sensor noise, lighting, weather changes
 - Style or domain differences
 - Time drift
- Effects
 - Model predictions worsen and model becomes less confident
 - Accuracy drops



Problem Statement

- ML models often fail when the test data distribution differs from training
- Retraining is impractical and expensive— requires more labeled data
- **How can we make models more robust to distribution shifts at test time?**



[1]

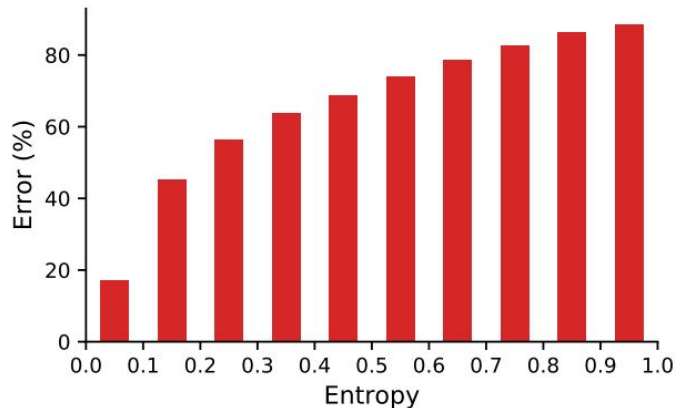


Research Question

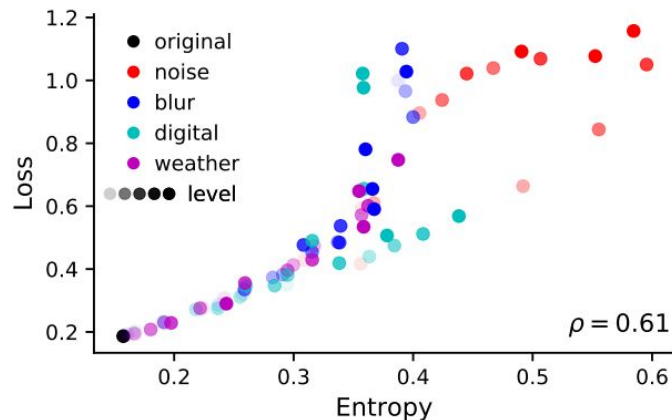
Can test-time entropy minimization improve model robustness to distribution shifts?

Base Paper - “TENT: FULLY TEST-TIME ADAPTATION BY ENTROPY MINIMIZATION” [1]

- TENT = Test Entropy minimization
 - Allows models to *adapt at test time* by minimizing prediction entropy
- Entropy = uncertainty in model predictions
 - Minimizing entropy makes outputs more confident and reliable under distribution shifts



[1]



[1]

Base Paper - Method

1. Train model normally on clean data
2. At test time, the model processes batches of new (possibly corrupted data)
3. For each batch:
 - a. Pass the batch through the network to get predictions
 - b. Compute entropy loss of the predictions
 - c. Compute gradients of entropy loss with BN parameters (scale/shift) using backprop
 - d. Update BN parameters for each channel to reduce entropy

$$H(\hat{y}) = - \sum_c p(\hat{y}_c) \log p(\hat{y}_c)$$

[1] - Shannon entropy formula

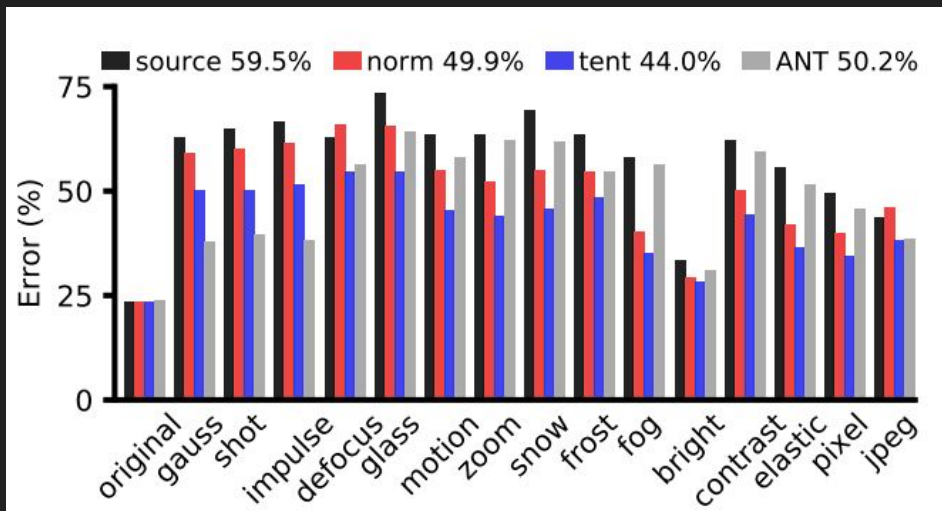
$$\begin{aligned} \text{normalization} \quad & \mu \leftarrow \mathbb{E}[x_t], \sigma^2 \leftarrow \mathbb{E}[(\mu - x_t)^2] \\ \text{transformation} \quad & \gamma \leftarrow \gamma + \partial H / \partial \gamma, \beta \leftarrow \beta + \partial H / \partial \beta \end{aligned}$$

[1] - BN parameters

Base Paper - Results

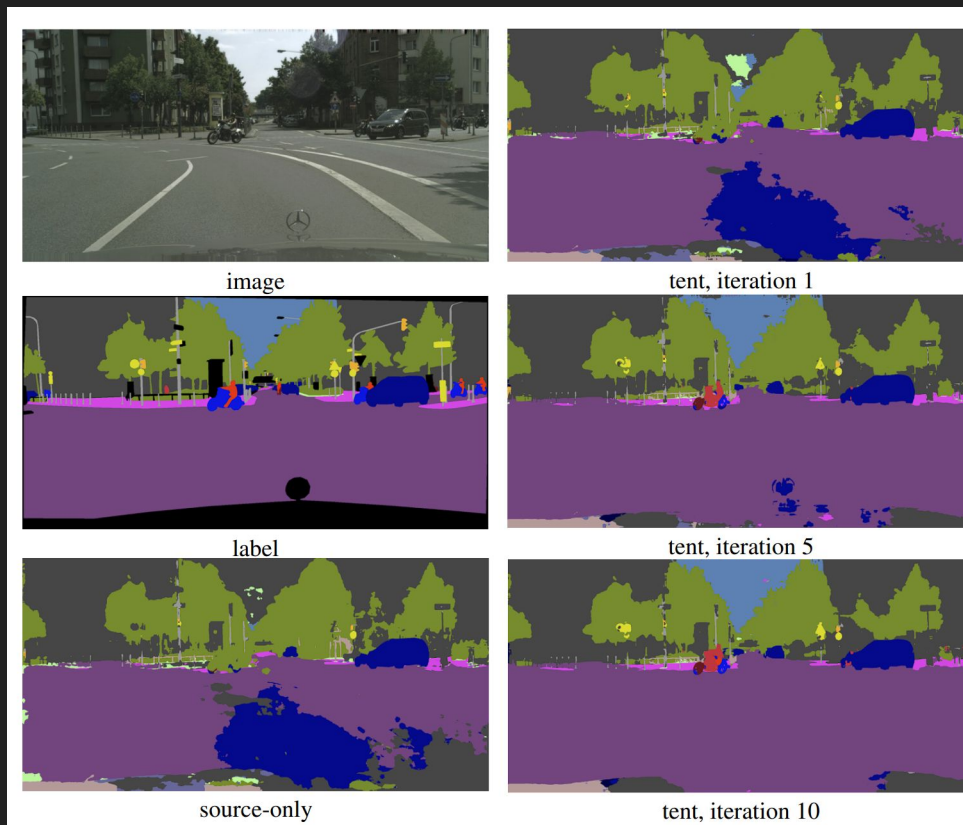
- Test-time adaptation is possible and effective
- Entropy minimization works as a general, unsupervised adaptation signal
- Updating only batch normalization parameters is sufficient

- “For robustness to corruptions, tent reaches 44.0% error on ImageNet-C, better than the state-of-the-art for robust training (50.2%) and the strong baseline of test-time normalization (49.9%).” [1]



[1] - Corruption error comparison

Base Paper - Results (cont.)



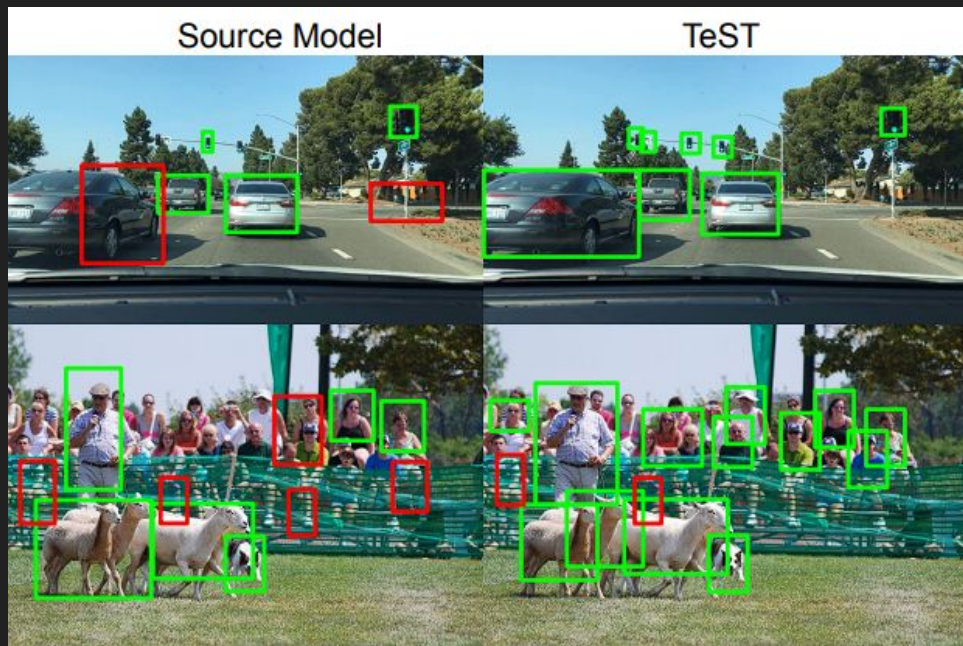
[1] - Adaptation for semantic segmentation with TENT

Related Papers: “Test-Time Adaptation to Distribution Shift by Confidence Maximization and Input Transformation” [2]

- Attempts to improve upon the issues of premature convergence and instability
- Replaces entropy with non-saturating surrogate loss
 - Avoids plateauing when entropy is already low, preventing premature convergence
- Adds diversity regularizer
 - Maximizes entropy across batch— avoids the model collapsing to one class for all samples
- Adds input transformation module
 - Trained at test time using unsupervised loss
 - Modifies each test image slightly to make it more like training data
 - Usually implemented as a small CNN or learned filter before the input

Related Papers: “TeST: Test-time Self-Training under Distribution Shift” [3]

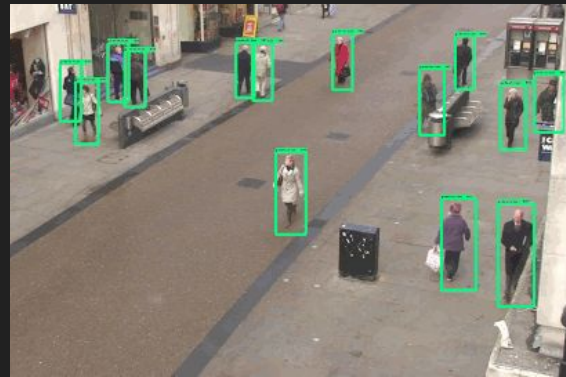
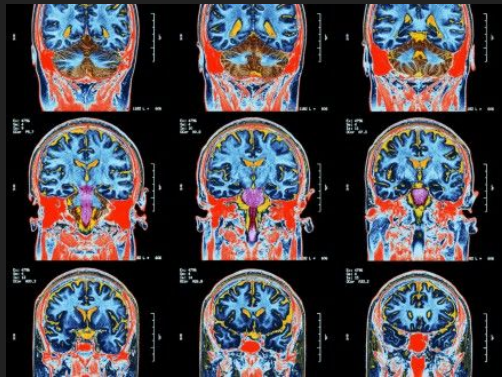
- Alternate approach to test-time adaptation
- Adapts the model using a student-teacher framework
- Teacher: original pretrained model, predicts on unlabeled test data
- Student: Model that learns from the teacher’s model confident predictions



[3] - Region-based CNN. Red = false positive, green = true positive.

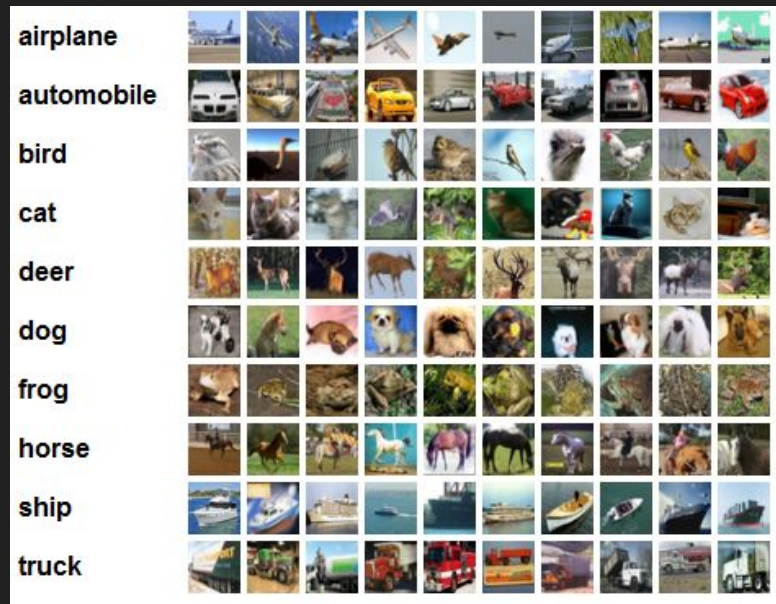
Trends and Impact

- Growing focus on test-time adaptation instead of retraining
- Real-world impacts:
 - Autonomous vehicles: adapt in real-time to new conditions (weather, lighting)
 - Medical imaging: Handle variations in medical imaging from scanners
 - Security: Remain accurate when environment shifts



Planned Experiment

- Train phase:
 - Train model on clean dataset (CIFAR-10)
- Test phase:
 - Corrupt the test data (add noise, blur, brightness, pixelation, fog, etc.) using CIFAR-10-C's corruptions
 - Implement TENT on corrupt data
- Evaluate:
 - Compare baseline (no adaptation) vs TENT adaptation on corrupted test sets
 - Measure changes in accuracy, entropy, and other metrics

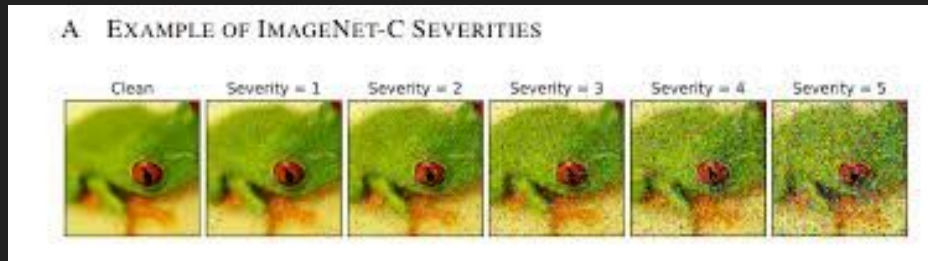


CIFAR-10 dataset, has 10 classes

Expected Results

- Baseline accuracy drops under distribution shift
- TENT adaptation improves accuracy
- TENT shows even stronger improvement under more severe corruption

Method	Source	Target	Error (%)	
			C10-C	C100-C
Source	train		40.8	67.2
RG	train	train	18.3	38.9
UDA-SS	train	train	16.7	47.0
TTT	train	test	17.5	45.0
BN		test	17.3	42.6
PL		test	15.7	41.2
Tent (ours)		test	14.3	37.3



[1] - highest severity corruption benchmark

Possible Extensions

- Use broader datasets like CIFAR-100 and ImageNet
- Measure runtime and efficiency
- Add lightweight preprocessor like in the second paper
- Corruption combinations
- Ablation study of parameters
- Experiment with different batch size (like single-instance)
- Adapt other parts of the network



CIFAR-100
dataset, has
100 classes

References

- [1] Wang, Dequan, et al. *Tent: Fully Test-Time Adaptation by Entropy Minimization*. ICLR, 2021.
- [2] Mummadi, CK, et al. *Test-Time Adaptation to Distribution Shift by Confidence Maximization and Input Transformation*. NeurIPS, 2021.
- [3] Sinha, Samarth. et al. *Test-Time Self-Training under Distribution Shift (TeST)*. CVPR, 2023.